

ValorAI

Explainable Real Estate Intelligence

"From Raw Data to Governed Valuation."

Arab Academy for Science & Technology & Maritime Transport
College of Artificial Intelligence – Alamein
Course: GN341 – Mobile Computing
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Presenter: Mohammed Hassan Mahmoud



Presentation Roadmap



Project Narrative: A systematic journey from real-world market opacity to a governed, spatial intelligence platform. We transition through data sanitization and mathematical filtering towards an agentic orchestration layer.

| The Real Estate Challenge in Egypt

01

Structural Market Opacity & Data Contamination

\$10B+ annual transaction volume unrecorded and fragmented. Portals polluted with duplicates, stale pricing, and placeholder values (≤ 100 EGP).

FRAGMENTED DATA

STALE LISTINGS

FAKE PRICING

02

Appraisal Latency & Subjective Variance

Manual appraisal workflows take 3–5 days per property. High result variance due to subjective judgment and lack of standardized spatial evidence.

3–5 DAY LATENCY

HUMAN BIAS

NON-STANDARDIZED

03

Algorithmic Failure in Standard LLMs

Standard models lack direct database grounding. Failure to perform precise arithmetic or spatial verification leads to confident price hallucinations.

HALLUCINATION RISK

ARITHMETIC FAILURE

UNCHECKED CONTEXT

ValorAI — Dual Architecture Platform



Fair Price Engine

- Automated Valuation Modeling (AVM)
- Spatial Retrieval & Geometric Indexing
- MAD Outlier Pruning & Similarity Decay
- Statistical Quantile Pricing (P20/P50/P80)



Grounded Copilot

- Agentic Negotiation Assistant
- "What-If" Scenario Simulation
- Evidence Aggregation & Explainability
- Deterministic Tool-Calling Pipeline

BRIDGE: GOVERNED EXECUTION LAYER

LLM restricted to presentation | All arithmetic offloaded to Python kernel

⚡ Engine Latency: < 15ms

🛡️ System State: **GOVERNED**

Precision Valuation + Governed Intelligence

End-to-End User Journey



i The journey transitions from structured mathematical valuation to flexible, governed agentic negotiation.

Premium UI/UX Design System



Visual Language

Material 3 Dark Theme base
HSL-tailored color harmonies
Smooth spatial gradients
Micro-interaction animations



Contextual Widgets

Category-Aware input logic
Residential / Commercial toggle
Buy / Rent workflow adaptation
Zero-placeholder empty states



Regionalization

Full RTL (Right-to-Left) support
Bilingual Arabic/English labels
Egyptian market terminology
Dynamic currency formatting



Dynamic Property Result Cards

Property Summary: Real-time core metrics
Confidence Score: Statistical reliability gauge

Evidence Layer: SHAP drivers & Comparables
Market Signals: Neighborhood trends

● PRECISION ● EXPLAINABILITY ● RESPONSIVENESS

MATERIAL 3 DESIGN SYSTEM

System Architecture & Integration Layers

LAYER 1



Client Layer

Flutter Mobile Application Interface

Dio HTTP Client

LAYER 2



Gateway Layer

FastAPI High-Performance API Gateway

JWT + OAuth2

LAYER 3



Valuation Engine

Fair Price Engine — Goldilocks Routing

• CMT Engine • ML Engine

Python Core

LAYER 4



Persistence Layer

PostgreSQL Relational Storage + Spatial Extensions

PostGIS + Alembic

LAYER 5



Orchestration Plane

Copilot Orchestrator & Truth Layer Governance

Agentic Control

Engineering Journey — From Scraper to Platform

P1: INGESTION

- Raw Data Scraping
- Regex Sanitization
- Database V1 (Flat)

P2: SPATIAL

- DB V2 (Hierarchy)
- DB V3 (Geometry)
- PostGIS Integration

P3: ENGINES

- Fair Price Engine
- CMT Engine (Spatial)
- ML Engine (CatBoost)

P4: ROUTER

- **Hybrid V1 (Failure)**
- **Hybrid V2 (Failure)**
- **Goldilocks Router V3**

P5: AGENTIC

- Explainability Layer
- Truth Layer / Tools
- Copilot Orchestrator
- Flutter Deployment

Full Evolution Chain:

Scraping → Cleaning → DB V1 → DB V2 → DB V3 → PostGIS → Fair Price Engine → CMT → ML → Hybrid V1 → Hybrid V2 → Goldilocks Router → Explainability → Truth Layer → Tools Layer → Copilot → Orchestrator → Flutter App

Ingestion Crawler & Sanitization Pipeline

NEXT_DATA Parser

Extracts JSON payloads from SSR HTML script tags.
Request Throttle: **1.2s** crawl interval.

Sanitization Metrics

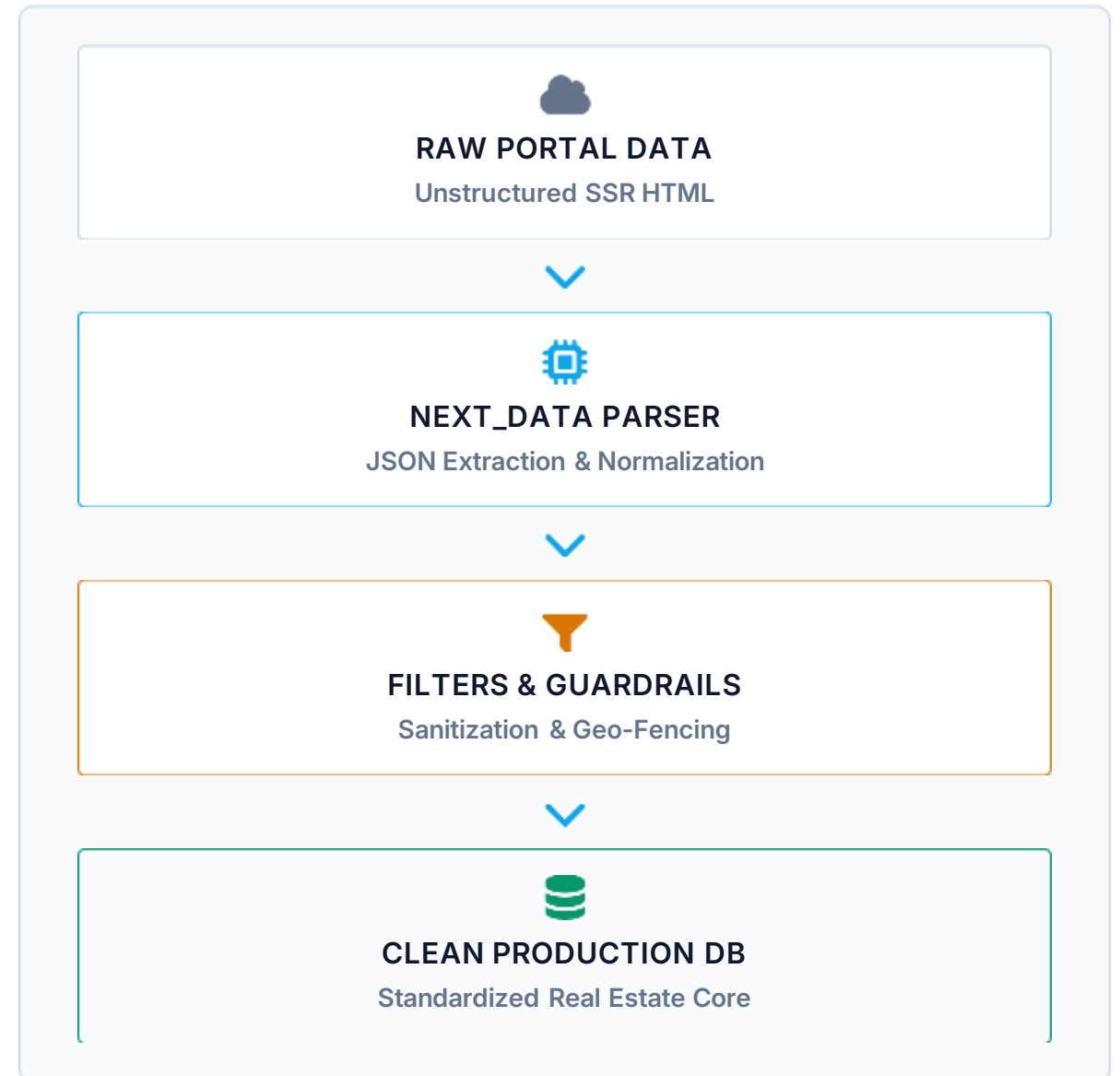
Pruned **20,490** raw rent records to
16,730 unique high-fidelity records.

Data Guardrails

Drops Price $\leq$ **100 EGP**.
Area Bounds: **5 – 10,000 sqm**.

Geographic Bounding Box

Latitude	[22.0, 32.5]
Longitude	[24.0, 36.0]



Database Schema Evolution — V1 → V3



INITIAL PROTOTYPE

V1: Flat Relational

- Flat property columns with string-based area fields
- Latitude/Longitude stored as FLOAT primitives
- ⚠ Spatial overlaps computed in Python ($O(N^2)$ complexity)
- ⚠ Redundant data storage; no area normalization



RELATIONAL STRUCTURE

V2: Hierarchical

- ✓ Parent-child adjacency table (parent_area_id)
- ✓ Strict Hierarchy: Governorate → District → Neighborhood
- Recursive CTE queries for spatial adjacency walks
- ⚠ Lacked native geometric containment verification



PRODUCTION SYSTEM

V3: PostGIS Spatial

- NativeMultiPolygongeometry for area boundaries
- Pointgeometry for listings (SRID 4326)
- GIST spatial indexing for sub-10ms retrieval
- Native ST_Covers() and ST_DWithin() integration



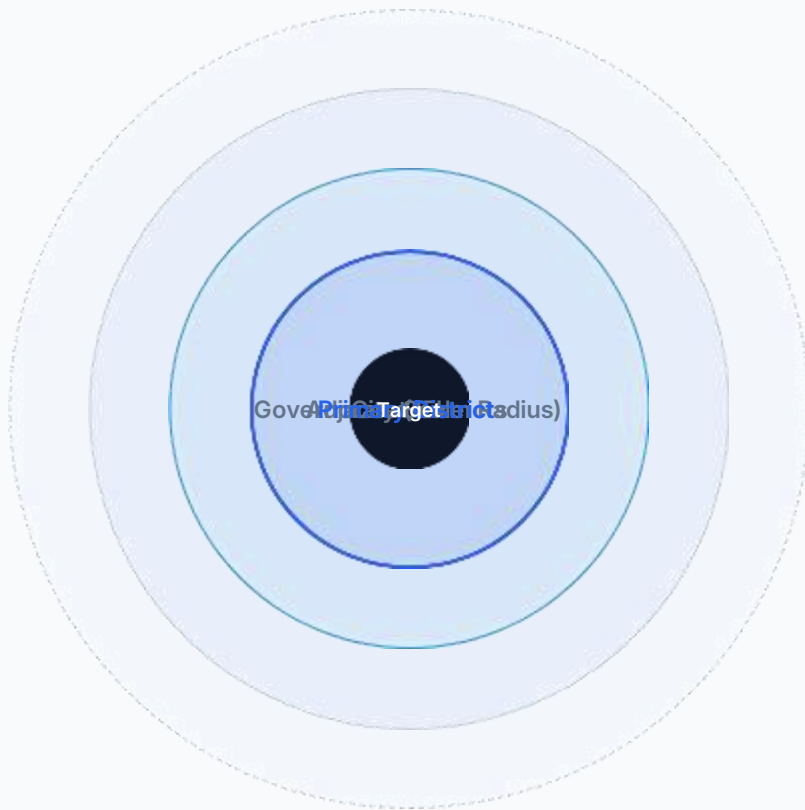
Alembic Migration

[c66d987da7fa](#)

Tenant-isolated tables for workspaces, chats, and property scenario states.

PostGIS Integration — Spatial Indexes & Geofencing

SPATIAL HIERARCHY RETRIEVAL MODEL



Spatial Operators

< 10ms

`ST_Covers(geom, Point)` resolves parent boundary containment.
`ST_DWithin` performs high-speed radial neighbor searches.

Indexing Architecture

GIST Index: Native geometry indexing for bounding box acceleration.
Composite B-tree: Category, Property Type, Bedrooms, and Area ID.

Address Resolution Cache

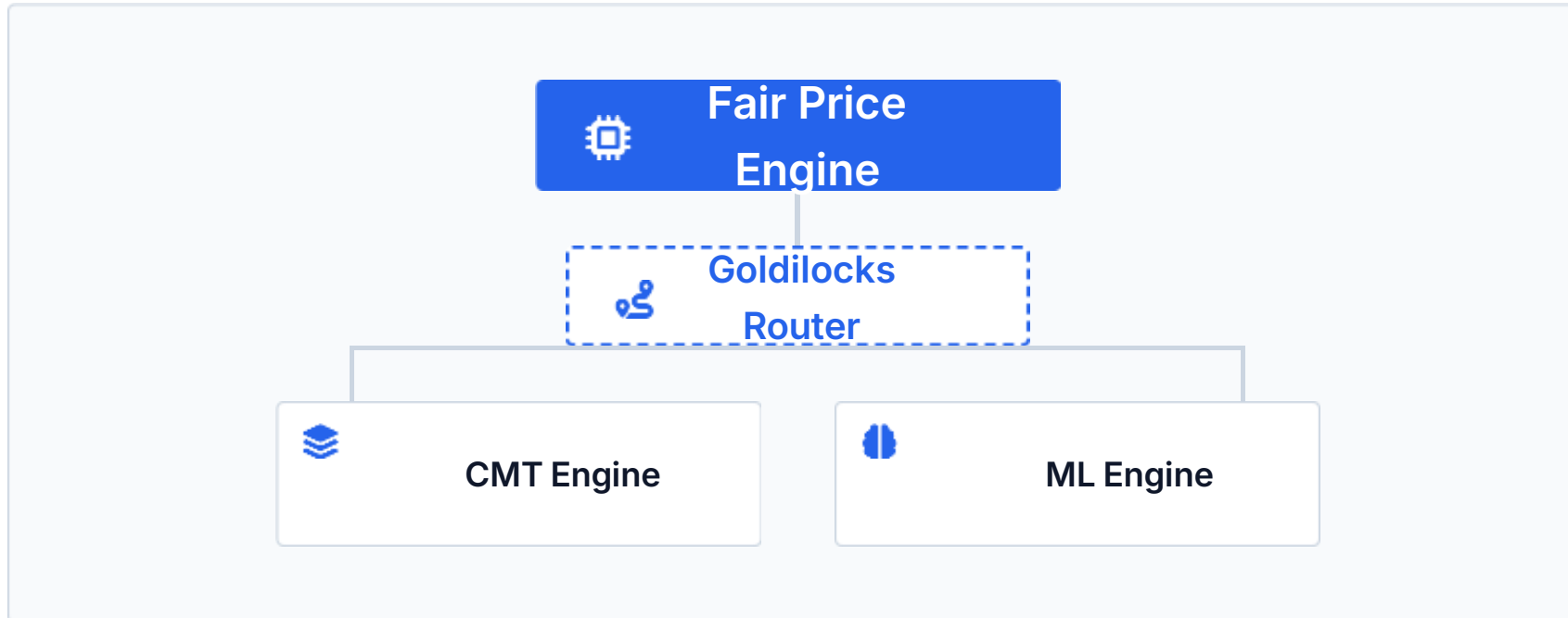
< 5ms

Normalized inputs matched against `address_resolution_cache`.
Eliminates redundant Google Maps API calls for known coordinates.

Adjacency Walks

Recursive CTE queries perform topological walks to select sister districts while strictly excluding primary parent overlaps.

Fair Price Engine — Master Valuation System

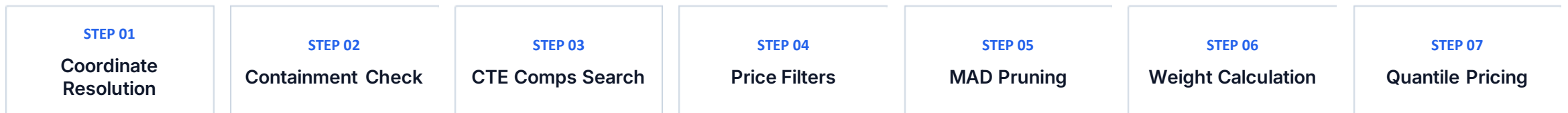


REST API ENDPOINT

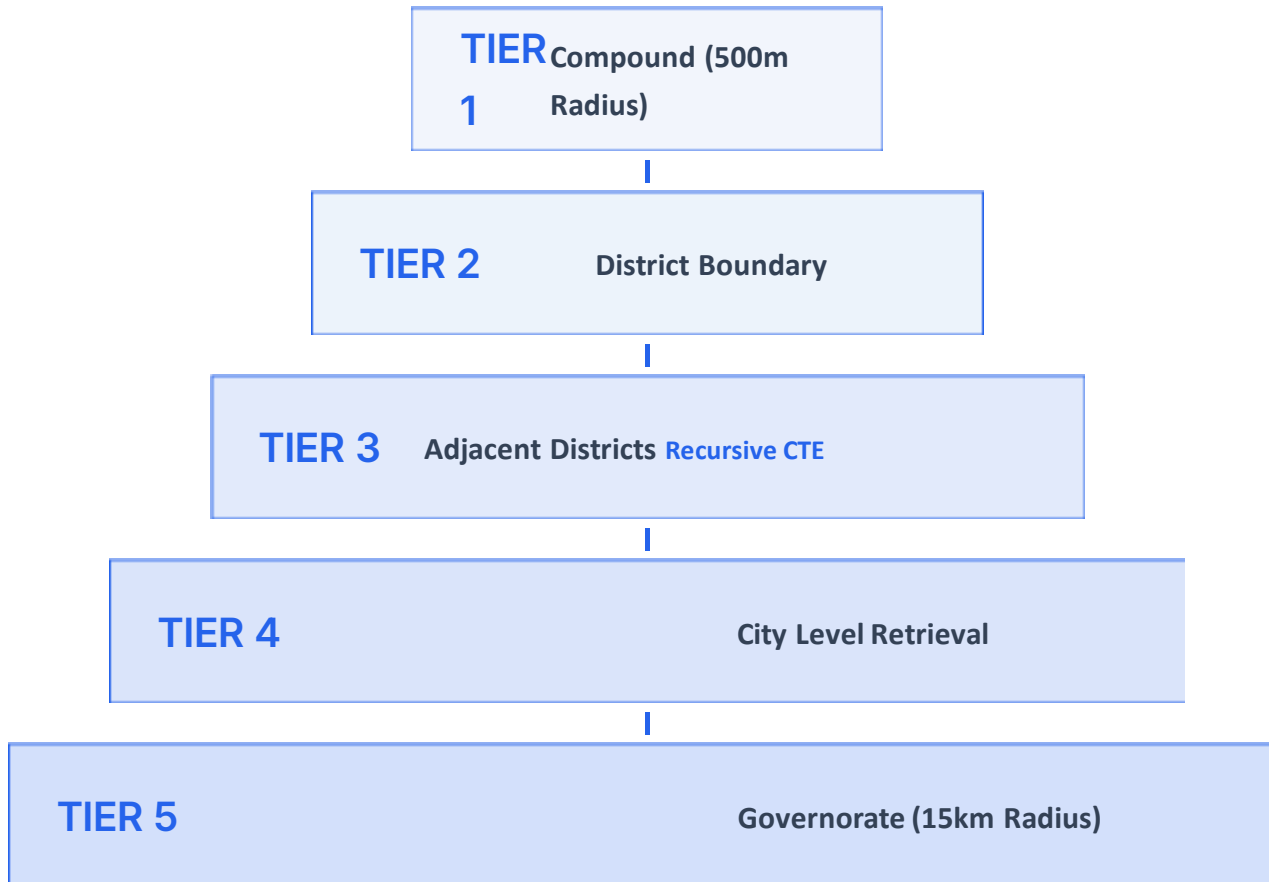
POST /v1/valuation/fair-price

Core Architecture: The Fair Price Engine is the authoritative parent system. It encapsulates routing logic and multiple valuation pipelines into a unified request handler.

★ Internal Stage Pipeline



CMT Engine — Comparable Market Technique



Spatial search expands dynamically until 40 comparable target is achieved.

🔹 Hard Constraint Filters

- ✅ **Core Match:** Category, Period, and Property Type must be identical.
- ✓ **Spatial Validity:** Point-in-Polygon containment via [ST_Covers](#).
- 💖 **Amenity Overlay:** Requires ≥ 1 matching amenity if data density allows.

⚙️ Deterministic Logic

- 🛡️ **No Probability:** Pure spatial statistics on observed neighbor listings.
- 🔗 **Recursive CTE:** Dynamic adjacency walk across district hierarchies.
- ⚡ **Performance:** Sub-10ms retrieval using composite GIST indexes.

CMT Core Mathematics

Deterministic Spatial Aggregation

$$\text{Price}_{\text{fair}} = \text{Median}_w(\{x_1, \dots, x_n\})$$

Where $w_i = f(\text{dist}, \text{size}, \text{age}, \text{amenities})$

The CMT engine prioritizes robustness over sensitivity. By utilizing a weighted median approach combined with non-parametric outlier detection, the system remains stable despite the high variance and "pollution" found in Egyptian real estate portal data.

MAD OUTLIER FILTER

$$MZ_i = 0.6745 \cdot (x_i - \text{med}) / \text{MAD}$$

Prunes listings where $|MZ_i| > 3.5$.
Resistant to extreme "placeholder" prices that distort mean-based models.

Robust Statistics

SIMILARITY DECAY

$$W_i = \prod (1 - \Delta_{\text{feature}} / \sigma)$$

Calculates weights based on Jaccard Index for amenities and Euclidean distance for spatial proximity.

Weighted Attribution

QUANTILE PRICING

$$\text{Range} = [P_{20}, P_{50}, P_{80}]$$

Provides a confidence interval rather than a point estimate. P50 represents the expected fair market value.

Confidence Bounds

Machine Learning Engine

🎯 Interpolation & Coverage

The CatBoost engine resolves cold-start compounds where direct comparable listings are unavailable. It provides valuation stability in both extremely sparse rural areas and hyper-dense urban clusters.

📊 Mathematical Foundation

TARGET TRANSFORMATION $y = \ln(\text{price} + 1)$

INFERENCE INVERSION $\text{Price}_{\text{EGP}} = e^y - 1$

📦 Feature Engineering

GPS Coordinates

Uber H3 Hexagons

Property Age

Amenity Scores

Category ID

Size (SQM)

Log-transformation protects the model from over-indexing on extreme luxury outliers and stabilizes variance across diverse price points.

🔧 Deployed Artifacts



rent.cbm

1.3MB • CatBoost



sale.cbm

2.2MB • CatBoost

Valuation Explainability & SHAP Drivers

ML Explainability (CatBoost)

SHAP IMPACT CONVERSION

$$\text{Impact_EGP} = \text{Price} \times (e - 1)$$

- ✓ **Feature-Level Drivers:** Individual impact of size, age, and location converted to currency.
- ✓ **Log-Space Inversion:** Protects against luxury variance while maintaining linear explainability.
- ✓ **Structured Narratives:** Automated generation of location precision confirmation reports.

CMT Evidence (Spatial)

-  **Comparable Listings:** Detailed audit trail showing distance (km) and similarity %.
-  **Reason Codes:** Explain why specific listings were matched (e.g., "Same Compound", "Recursive CTE").
-  **Outlier Suppression:** Transparent logging of listings excluded by MAD filters.

Containment Precision

ST_Covers

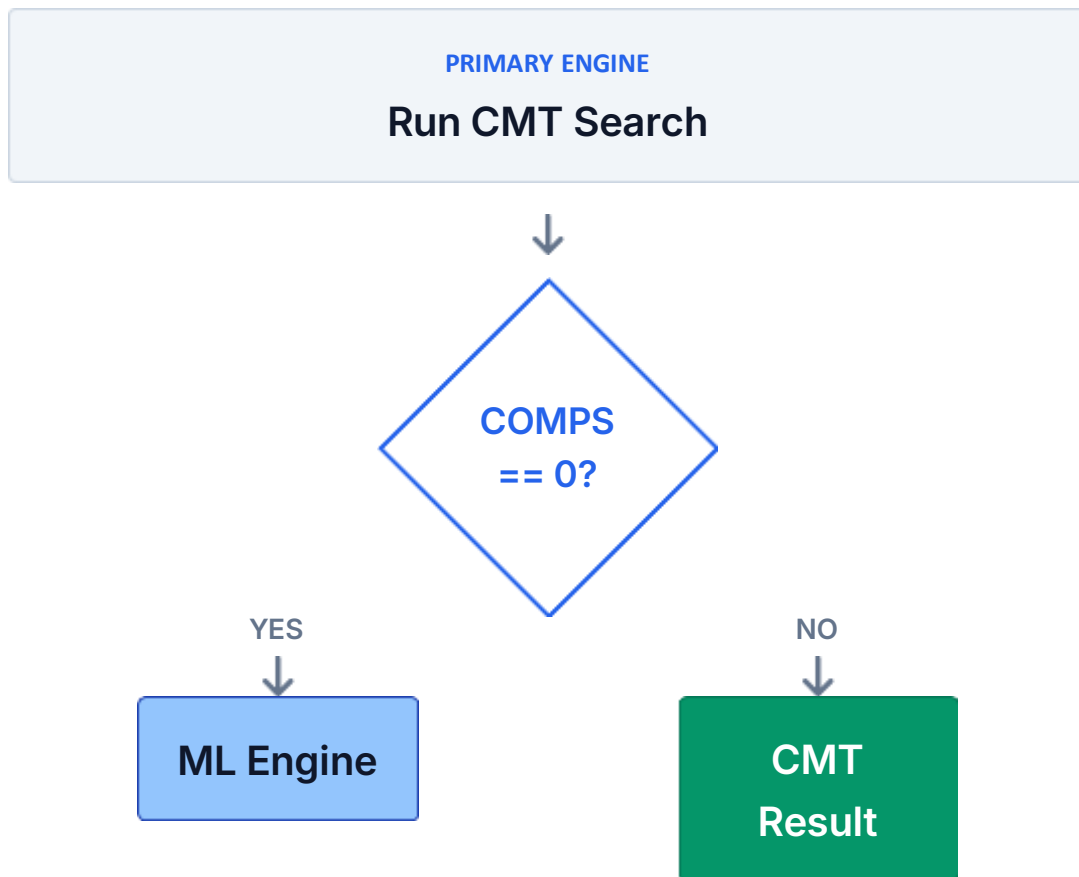
Spatial Radius

0.5 - 15km

 **Core Principle:** No black boxes — every estimate is mathematically and spatially explainable.

Hybrid V1 — Direct Fallback Failure

INTENDED ENGINEERING LOGIC



OBSERVED FAILURE MODE

⚠ The Sparse Data Trap

Case: Property in remote Giza desert area.

CMT Search Radius 15.0 km

Listings Found 1 (Stale/Placeholder)

Router Logic 1 != 0 → Stay in CMT

RESULTING VALUATION

12,500,000 EGP

(Based on a single unverified listing)

- ✗ Zero-check is insufficient for data quality.
- ✗ Sparse areas produce high-variance CMT noise.

💡 Engineering Lesson: Router must trigger fallback **BEFORE** complete search failure (Threshold > 0).

Hybrid V2 — Simple Density Threshold Failure

INTENDED ROUTING LOGIC

```
IF comps_count ≥ 10:  
    USE CMT_ENGINE  
ELSE:  
    USE ML_ENGINE
```

Simple binary threshold designed to ensure statistical significance for Comparable Market Technique before execution.

⚡ LATENCY CRITICAL

High-Density Latency Spike

In districts like Madinaty (>700 listings), sorting and similarity scoring in Python caused a 300ms bottleneck.

🔊 SIGNAL NOISE

Noisy Estimates in Dense Hubs

CMT became oversensitive to local price variations in high-density areas, failing to provide a stable "fair" baseline.

⚠️ Density alone is an insufficient routing signal.

⚠️ Both sparsity AND extreme density are failure modes.

Hybrid V3 — The Goldilocks Router

<
15ms

SYSTEM LATENCY

⚡ DETERMINISTIC SPEED

ROUTER DECISION MATRIX

ML Engine	CMT ENGINE		ML Engine
0 Comps	11 Comps	50 Comps	Max
Goldilocks Zone (11–50) Optimal balance for CMT. Statistical significance meets high-performance Python sorting. VALUATION_OK		Edge-Case Logic Overrides to CMT for unseen compounds if ≥ 10 comps found in district. Prevents cold-start failures.	
Shadow Execution Alternate engine runs in BackgroundTasks. Drift detection identifies model divergence without blocking users.		Performance Guard ML handles high-density (>50) to avoid CMT sorting latency spikes. Ensures sub-15ms SLA.	

"Not too sparse. Not too dense. Just right."

Modular Tools Layer — 8 Functional Tools

BUSINESS LOGIC ENCAPSULATION

T1  Valuation Resolves precise pricing estimates and confidence scores.	T2  Explainability Compiles SHAP feature driver lists and logic narratives.	T3  Comparables Retrieves nearest neighbor listings and spatial evidence.	T4  Fairness Audits target offer prices against computed range bounds.
T5  What-If Simulates property pricing deltas via detail edits.	T6  Negotiation Generates grounded counter-offer talking points.	T7  Investment Evaluates investment risk scores and yield signals.	T8  Market Insight Calculates workspace-wide aggregation statistics.

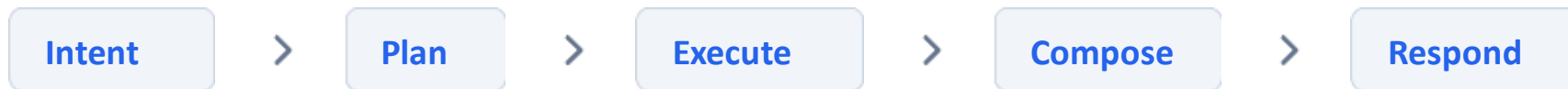


LLM has ZERO direct SQL access — Only tool execution permitted.

BUSINESS_LOGIC_ISOLATION_v1.0 // TENANT_JWT_BOUNDARY_ENFORCED

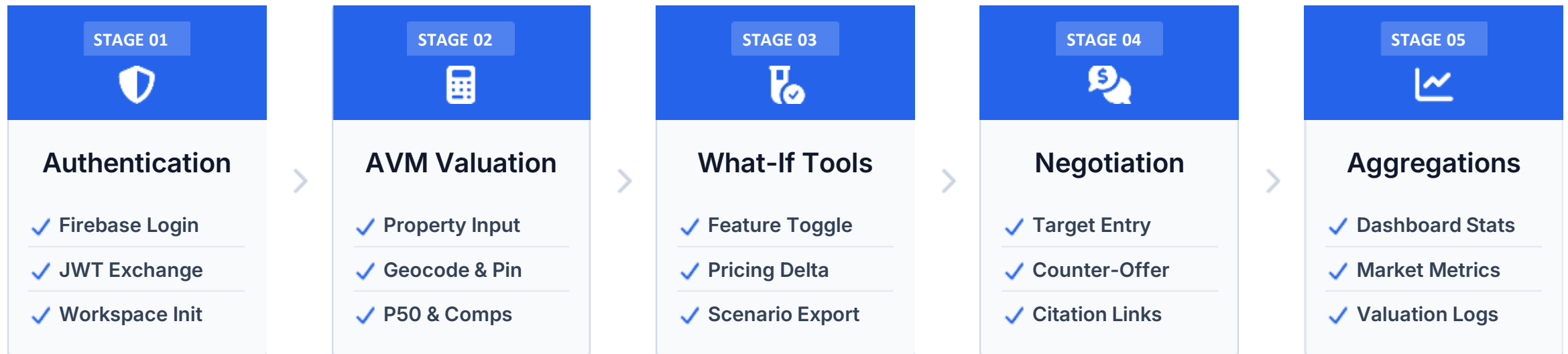
GOVERNED EXECUTION

Copilot Orchestrator — Control Plane



Control Plane Isolation: Tenant boundaries enforced via JWT exchange; LLM has zero direct SQL access.

Live Demonstration — 5-Stage Sequence



Full-Stack Integration State

Flutter App v1.0

FastAPI Gateway

PostGIS Database

Truth Layer Active

Technical Achievements — Empirical Validation

20.00%

MEDIAN ABSOLUTE PERCENTAGE ERROR
(MDAPE)

0.7444

R² Coefficient

80.60%

Coverage Score

EVALUATION FRAMEWORK

Leave-One-Out (LOO) test conducted on 500 records across the deterministic CMT pipeline. Validates the hybrid engine's ability to generalize spatial pricing.

📍 Cairo-Alex Desert Road	8.33% MdAPE
📍 Al Rehab City	12.50% MdAPE
📍 Al Narges	27.27% MdAPE
📍 4th District (Sparse)	72.67% MdAPE



CMT + ML Architecture guarantees full market coverage.

Future Engineering Priorities

PHASE 01



Automated Retraining

- Monthly CatBoost pipeline integration
- Automated hyperparameter tuning
- Model drift detection triggers

PHASE 02



Polygon Boundaries

- Native ST_Intersects logic
- Replace circular radial buffers
- True neighborhood containment

PHASE 03



Smart Autocomplete

- Production-ready location search
- Coordinate-aware address caching
- Optimized geocoding latency

PHASE 04



Market Expansion

- Multi-city coverage (Alex/Giza)
- Enterprise tenant workspaces
- Real-time telemetry analytics

Strategic Focus: Transitioning from a validated prototype to a production-grade, self-evolving spatial intelligence platform.

Thank You

Project Contributions



Spatial Intelligence Foundation

Engineered a robust PostGIS infrastructure for sub-10ms native geographic querying and geofencing.



Hybrid Valuation Architecture

Developed the Goldilocks Router to optimize precision across sparse and dense market segments.



Governed AI Orchestration

Implemented a Truth Layer and Tooling boundary to ensure grounded, hallucination-free intelligence.



Questions?

Opening the floor for discussion
regarding the ValorAI defense
presentation.